

- LAB 45: Light & Reflection Investigation - Data Sheet
 - TEST A: Law of Reflection — Angle Measurements
 - TEST B: Obstacle Course Results
 - TEST C: Surface Comparison
 - LAB QUESTIONS & EXIT TICKET

LAB 45: Light & Reflection Investigation - Data Sheet

Name(s): _____ Date: _____

Mission: Prove the Law of Reflection and engineer a multi-mirror light path to bypass an obstacle and hit a target!

TEST A: Law of Reflection — Angle Measurements

Does Angle In = Angle Out? Measure from the normal line.

Angle In (Set)	T1 Angle Out	T2 Angle Out	T3 Angle Out	Avg Angle Out	Match? (Yes/No)
30°					
45°					
60°					

TEST B: Obstacle Course Results

Engineer a double-bounce path around the textbook.

Metric	Your Result
Attempts to hit bullseye	
Total beam path distance (cm)	

Metric	Your Result
Mirror 1 angle used	
Mirror 2 angle used	

TEST C: Surface Comparison

Which surface reflects the best?

Surface	Brightness (1-5)	Beam Type (Sharp/Fuzzy/None)
Glass Mirror		
Aluminum Foil		
White Paper		

LAB QUESTIONS & EXIT TICKET

1. Briefly explain: Why did the mirror reflect a sharp beam while the white paper did not reflect any beam at all?

2. Draw a simple diagram below showing how your Test B double-bounce solution worked. Label the flashlight, Mirror 1, Mirror 2, and the target.

(Draw your diagram here)

3. A submarine uses a periscope to see above the water. Based on your experiment, at what angle must each mirror inside the periscope be set to beam light straight down the tube? Explain why using the Law of Reflection.
